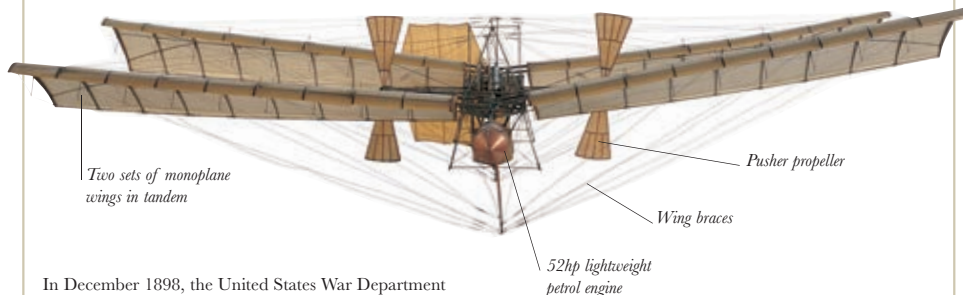


Langley Aerodrome A



In December 1898, the United States War Department commissioned Smithsonian Secretary Samuel Langley to build a man-carrying aircraft. Five years later, the Aerodrome A was ready for testing. The large tandem-wing machine was to be catapult-launched from a houseboat floating on the Potomac River. In both tests, in October and December 1903, the craft crashed straight into the river, the pilot surviving unhurt. The project was abandoned; it is now generally agreed that the design could never have flown.

ENGINE 52hp 5-cylinder radial petrol engine

WINGSPAN 48ft 5in (14.7m) **LENGTH** 52ft 5in (16m)

SPEED Unknown **CREW** 1

PASSENGERS None

R.E.P 1 (1907)

In 1904, French engineer Robert Esnault-Pelterie built a Wright-style biplane glider that was controlled by the first use of ailerons. When he turned to powered craft, his R.E.P 1 – a birdlike monoplane with a tapered wing – used wing-warping, although it did boast an innovative flight control stick. The excellent engine was his own design. The airplane made a number of short flights at Buc during November and December 1907. Later designs were more successful.

ENGINE 30-hp R.E.P 7-cylinder radial

WINGSPAN 31ft 5in (9.6m) **LENGTH** 22ft 3in (6.9m)

TOP SPEED Unknown **CREW** 1

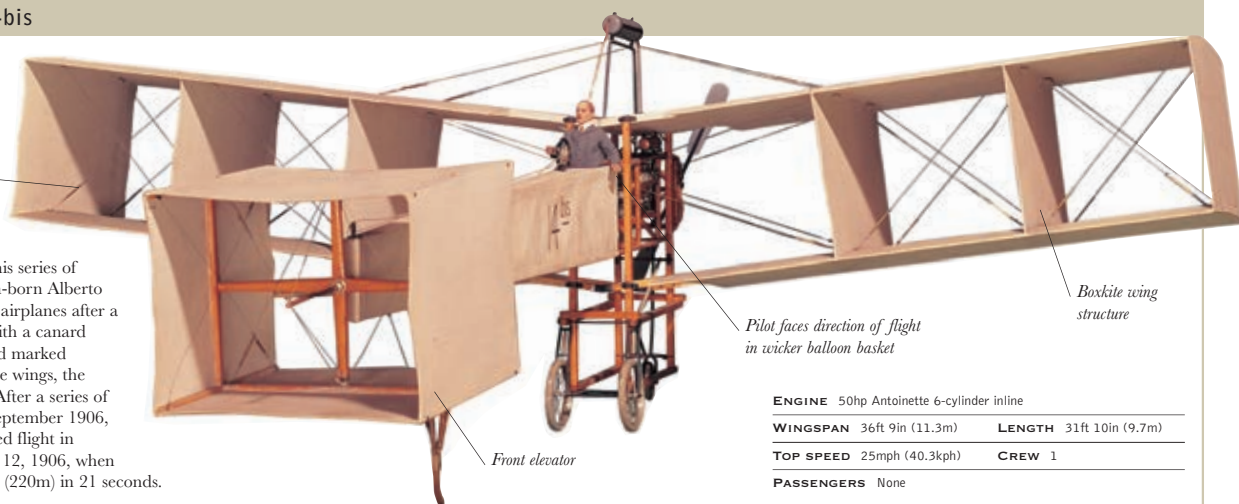
PASSENGERS None



Santos-Dumont 14bis

*High-tension
bracing wires keep
wing structure firm*

Renowned in Europe for his series of dirigible airships, Brazilian-born Alberto Santos-Dumont turned to airplanes after a visit to the US in 1904. With a canard (tail-first) configuration and marked dihedral on the box-kitelike wings, the 14bis was a unique sight. After a series of short hops beginning in September 1906, the first officially recognized flight in Europe was on November 12, 1906, when Santos-Dumont flew 722ft (220m) in 21 seconds.



ENGINE 50hp Antoinette 6-cylinder inline

WINGSPAN 36ft 9in (11.3m) **LENGTH** 31ft 10in (9.7m)

TOP SPEED 25mph (40.3kph) **CREW** 1

PASSENGERS None

Dorand 1910 Biplane

Captain-Engineer Jean Dorand began his study of heavier-than-air flight for the army in 1894. In 1910, he built the biplan-laboratoire using the Maurice Farman plan of forward elevators, staggered wings, tractor propeller, and biplane tail. Dorand used this aircraft as his test model; it included equipment for measuring speed, pitch, and roll, as well as a camera.

ENGINE 60hp air-cooled Renault

WINGSPAN 39ft 4in (12m) **LENGTH** Unknown

TOP SPEED Unknown **CREW** 2

PASSENGERS None



Wright 1905 Flyer III



The Wright Flyer III was the world's first practical powered airplane. The Wrights began testing in June 1905 and their final flight on October 5 covered over 24 miles (39km). In 1908, the Flyer III was modified, and given a second seat. In 1908, it flew again briefly at Kitty Hawk.

ENGINE 20hp Wright 4-cylinder water-cooled inline

WINGSPAN 40ft 6in (12.3m) **LENGTH** 28ft (8.5m)

TOP SPEED c.35mph (c.56kph) **CREW** 1

PASSENGERS 1 (from 1908)

Wright 1902 Glider (No.3)

With their third glider, the Wrights finally achieved their goal of controlled flight. Nearly 1,000 flights were made at Kill Devil Hills by the end of October 1902, with a top distance of 622ft (190m) and duration of 26 seconds achieved during these tests.

ENGINE None

WINGSPAN 32ft 1in (9.8m) **LENGTH** 16ft 1in (4.9m)

TOP SPEED Unknown **CREW** 1

PASSENGERS None

